

WEEK 7 OVERVIEW

An aerial photograph of a residential neighborhood that has been completely inundated with floodwater. The water is a murky, brownish-grey color, covering the streets, yards, and surrounding areas. Numerous houses with various roof colors (brown, grey, red) are visible, partially submerged. Green trees are scattered throughout the landscape, some standing in the water. A central road runs vertically through the middle of the image, with a few vehicles visible in the distance. The overall scene depicts a severe flooding event.

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CEVE 543

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TRENDS AT NEIGHBORING TEXAS STATIONS

i Note

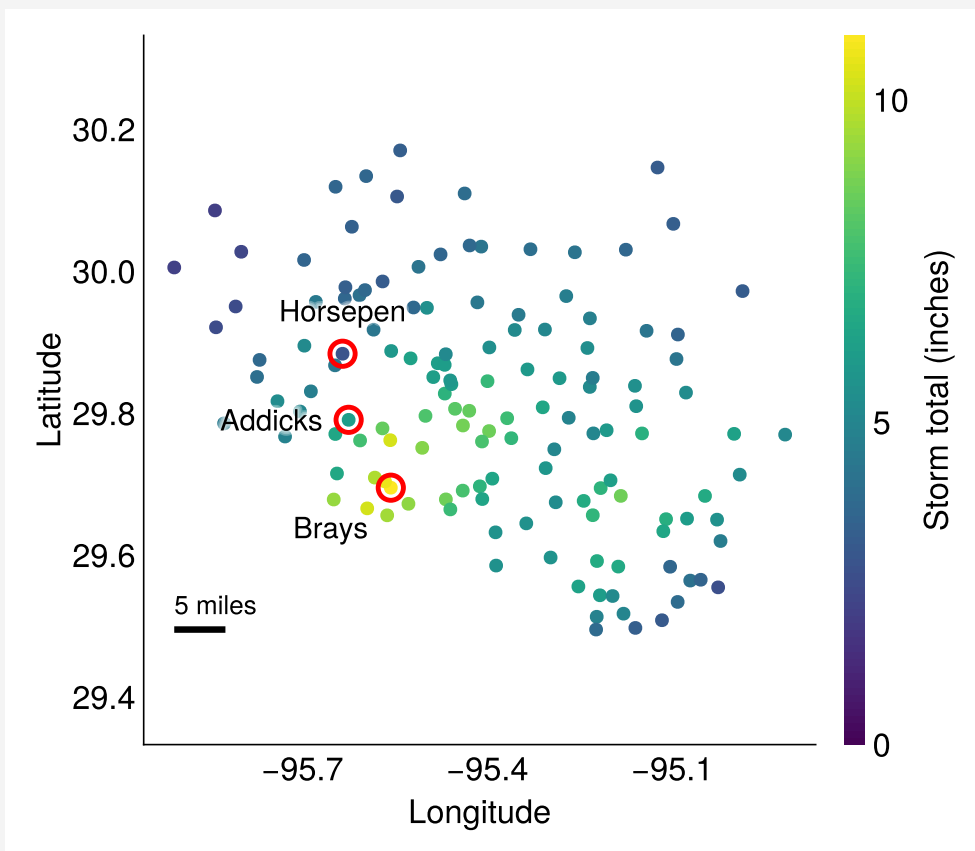
Figure shown in class: Kendall's rank correlation between global temperature and 1-hour (left) and 1-day (right) annual maximum rainfall at Texas stations; blue is positive and red negative (preliminary).

CO2 TRENDS FIT STATION BY STATION

i Note

Figure shown in class: The coefficient on CO2 in the location (top) and scale (bottom) of a nonstationary extreme value distribution, fit separately at each of 181 Gulf Coast stations; purple is an increase with CO2 and orange a decrease.

MEMORIAL DAY 2015

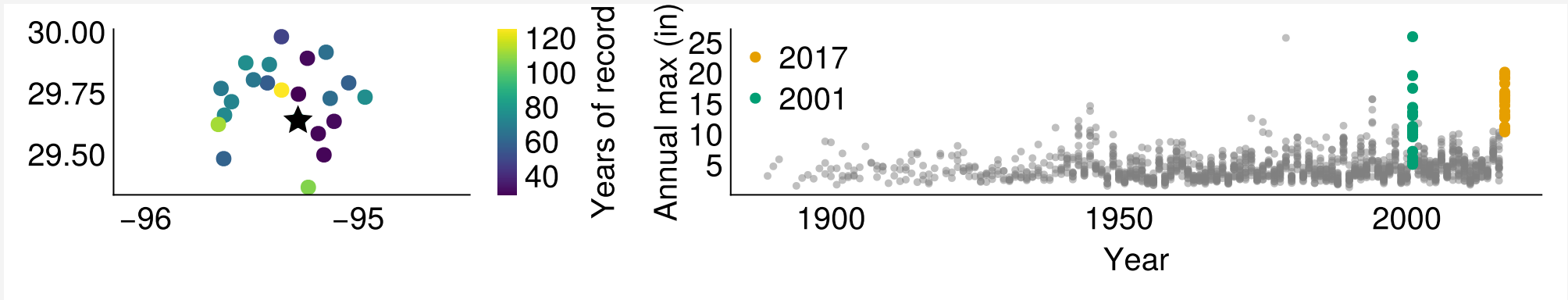


Totals for May 25-26 at 138 Harris County Flood Control District gauges.

- Brays Bayou: 11.00 inches.
- Addicks Dam, 7.8 miles from Brays: 6.08 inches.
- Horsepen Creek, 13.9 miles from Brays: 2.84 inches.

Within 14 miles, one storm's totals ranged from 2.84 to 11.00 inches.

ANNUAL MAXIMUM RAINFALL NEAR HOBBY AIRPORT



- 22 Atlas 14 stations within 25 miles of Hobby Airport (star) (Perica et al., 2018).
- 16 had their largest year in 2017 (Harvey), and 4 in 2001 (Allison).
- Fit alone, 100-year estimates run from 12.2 to 37.4 inches; the five largest come from 29- to 31-year records.

OBJECTIVES

If nearby stations see the same storms, pooling their records should average out where each storm landed.

By the end of this week, you can:

1. Describe the sources of noise in one station's record of extremes.
2. Describe, physically and statistically, the bias-variance trade-off in regionalization.
3. State at least three ways to regionalize and the key assumptions each makes.
4. Read a regional extremes analysis and assess whether its key assumptions hold.